

agent: discovery-architecture-design-agent cli: Claude Code CLI llm: claude-sonnet-4-6 run_id: 20260804T162417_gmxhab generated_at: 2026-08-04T10:56:13.000Z

1. Architecture & Design Hotspots Analysis

Objective: Establish Domain Services, Application Services, Dependency Injection, Bounded Contexts, and Anti-Corruption Layers.

Date: 2026-08-04 10:56:13 UTC | **Scope:** shende-shweta/pingcrm — PHP 8.2 / Laravel 11 (backend) · React 19 + TypeScript + Inertia.js (frontend)

Executive Summary

EXECUTIVE SUMMARY

PingCRM is a Laravel 11 + React 19 full-stack application built originally as a simple CRM demo, but now carrying a large IVR/telephony enterprise layer grafted on top. The backend is dominated by 82 single-action IVR controllers averaging 759 LOC each — each housing 55 `legacyEndpointN` methods that instantiate `GodService` classes via `new` (bypassing DI) and execute raw SQL strings in the same method as HTTP request handling. Twelve `Legacy/Services/*GodService.php` classes each define 50+ identically-structured workflow methods with hardcoded secrets, blocking `sleep()` calls, and mutable global-state caches. The frontend mirrors this with 133 `LegacyPass2_*.tsx` stub components (392 LOC each) and 124 legacy hooks that hard-code `fetch()` calls directly against private IVR endpoints with no shared API layer. The two dominant risks are **change amplification** — any schema or workflow change must be replicated across 55+ endpoint methods in each of 82 controllers — and **hidden coupling** — the IVR dashboard controller joins CRM `organizations` tables directly with IVR call-record tables, making the two domains inseparable at the database level.

91

CONTROLLERS / HANDLERS

16

MODELS / ENTITIES

12

SERVICE CLASSES FOUND

12

REPOSITORY CLASSES
FOUND

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by High-Risk Fat Controllers (H1, avg 722 LOC), God Classes (H7, 12 GodServices + 82 controllers at 759 LOC), Direct SQL in Controllers (H6, 88% non-compliant), Missing Repository Pattern (H3, 80 direct DB access points), Shared Database Coupling (H9), Domain Boundary Violations (H8), and Legacy Component Patterns (F5, 257 legacy artifacts).

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	~722 LOC (82 IVR controllers avg 759 LOC + 9 CRM controllers avg ~115 LOC)	HIGH RIS
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	82+ IVR controllers call <code>new GodService()</code> bypassing DI; CRM controllers use Eloquent inline without a service tier	HIGH RIS
H3	Missing Repository Pattern	Direct DB access points outside repositories	<10	10–20	>20	80 controllers with raw <code>DB::select / DB::table</code> ; 12 IVR Models also contain direct SQL in accessor methods	HIGH RIS
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 circular import cycles detected	GOOD
H5	Shared Utility Abuse	Utility files holding business logic	0	1–5	>5	5 Legacy Helper files + <code>IvrAccountContext</code> with DB query methods = 6 files	HIGH RIS
H6	Direct SQL in Controllers	ORM compliance % (queries kept out of controllers)	>90%	60–90%	<60%	~12% compliant — 80/91 controllers use raw <code>DB::select</code> or <code>DB::table</code> inline	HIGH RIS
H7	God Classes	Classes >1000 LOC (or >300 LOC with high method count)	0	1–3	>3	12 GodServices (373 LOC, 55 methods) + 82 IVR controllers (759 LOC, 57 methods) + 12 IVR Models (200+ LOC, 35 accessor methods) = 106 qualifying classes	HIGH RIS
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	<code>IvrHubController</code> joins CRM <code>organizations</code> in 2 private methods; <code>IvrAccountContext</code> imports <code>Organization</code> Eloquent model from CRM namespace = 7+ violations	HIGH RIS
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	47 IVR module tables + CRM tables in one DB; <code>organizations</code> table directly read by both IVR and CRM domains (~100% coupling)	HIGH RIS

F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	Weighted avg ~256 LOC (133 LegacyPass2 at 392 LOC; regular pages 55–141 LOC)	MODERATE
F2	Missing Frontend Service/Data Layer	Components with inline API calls	<10	10–20	>20	124 legacy hooks hard-coding <code>fetch('/ivr-legacy/...')</code> with no shared API client	HIGH RISK
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0 components strictly >400 LOC (LegacyPass2 peak at 392 LOC; IvrHubCharts at 141 LOC)	GOOD
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	Max 2 levels; Inertia server-passes data; no oversized global store detected	GOOD
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	133 LegacyPass2 static stub components + 124 legacy hooks with bare <code>fetch()</code> = 257 legacy artifacts	HIGH RISK

No additional hotspots beyond the standard set were observed. (The `extract($payload)` pattern and hardcoded `$apiKey` in GodServices are security risks addressed in the Security agent's report. The `$guarded = []` mass-assignment issue is likewise covered there.)

1.2 Hotspot-by-Hotspot Evidence

H1. Fat Controllers CRITICAL

Benchmark: Average LOC per controller = ~722 LOC → falls in the **High Risk** band (Good <150 · Moderate 150–300 · High Risk >300).

What to check: Business logic inside controllers/handlers — controllers should only translate HTTP ↔ application calls.

Evidence:

`app/Http/Controllers/Ivr/QueueManagementIndexController.php:1–759`

```
class QueueManagementIndexController extends Controller
{
    private $tenantId = 1; // hard-coded tenant – multi-tenant broken

    public function handleIndex(Request $request)
    {
        // Fat controller – business rules live here
        $service = new QueueManagementGodService();
        $q = $request->get("q");
        if ($q) {
            $rows = DB::select("select * from ivr_queue_managements where name
```

```

like '%" . $q . "%' and tenant_id = " . $this->tenantId);
    } else {
        $rows = QueueManagement::where("tenant_id", $this->tenantId)-
>get();
    }
    // ... legacyEndpoint1 through legacyEndpoint55 – same instantiation
    pattern
    }
}

```

This 759-LOC controller combines HTTP routing, SQL filtering, service instantiation, and 55 nearly-identical legacy workflow endpoints (legacyEndpoint1–legacyEndpoint55). All 82 IVR single-action controllers follow this identical blueprint for a total of ~62,238 LOC of controller fat.

app/Http/Controllers/Ivr/IvrHubController.php:1–381

```

class IvrHubController extends Controller
{
    private function loadStats(IvrAccountContext $ctx, array $filters): array
    {
        $queueQuery = DB::table('ivr_operational_queues')->where('account_id',
$ctx->accountId);
        $agentsQuery = DB::table('ivr_agents')->where('account_id', $ctx-
>accountId);
        $callsQuery = DB::table('ivr_call_records')->where('account_id', $ctx-
>accountId)->whereDate('started_at', $filters['date']);
        // ... 6 KPI calculations inline
        return ['active_calls' => $onCallAgents + $queued, 'queued_calls' =>
$queued, /* 4 more */ ];
    }
    // 8 more private DB-query methods: loadHourlyVolume, loadDailyTrend,
loadQueueDistribution...
}

```

381 LOC, 11 methods, every private method performs its own DB queries and result mapping — pure business logic embedded in the HTTP handler.

Why it matters here: Any change to IVR workflow logic (e.g. adding a new tenant filter, changing a column name) must be applied across up to 55 endpoint methods × 82 controllers = 4,510 places. **IvrHubController** cannot share its analytics calculations with a CLI command or a queued job because the logic is inlined alongside the HTTP response call.

Recommended approach:

1. Extract a thin **IvrWorkflowService** per module (e.g. **QueueManagementWorkflowService**) with a single **execute(WorkflowCommand \$cmd): WorkflowResult** method, inject via constructor.
2. Collapse the 55 **legacyEndpointN** methods into **legacyDispatch(Request \$request, int \$step)** that delegates to the injected service.

3. Move all `IvrHubController` private query methods to a `IvrDashboardQueryService` bound in the IoC container.
4. Enforce ≤ 150 LOC per controller with a `larastan/larastan` custom rule.

80 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109,	Fat controller — business logic + 55	Extract workflow logic to Application Service; collapse

[illegible]

[illegible]

	122, 135, 148, 161 +45	legacy endpoints inline	legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryExportController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryImportController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryIndexController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryStoreController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventorySyncController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern

<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109,	Fat controller — business logic + 55	Extract workflow logic to Application Service; collapse

	122, 135, 148, 161 +45	legacy endpoints inline	legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	19, 22, 44, 57, 70, 83, 96, 109, 122, 135, 148, 161 +45	Fat controller — business logic + 55 legacy endpoints inline	Extract workflow logic to Application Service; collapse legacyEndpointN to dispatch pattern

H2. Missing Service Layer CRITICAL

Benchmark: Controllers directly calling new GodService() = 82 IVR controllers → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: Business rules spread across controllers with no dedicated service tier.

Evidence:

`app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:20–30`

```
public function handleStore(Request $request)
{
    $service = new CallAnalyticsGodService(); // direct instantiation – no DI
    $q = $request->get("q");
    if ($q) {
```

```

        $rows = DB::select("select * from ivr_call_analyticss ...");
    } else {
        $rows = CallAnalytics::where("tenant_id", $this->tenantId)->get();
    }

```

Every IVR controller creates its domain service via `new XxxGodService()` inside the action method. Laravel's IoC container is never used, making tests impossible without monkey-patching.

app/Http/Controllers/ContactsController.php:32–62

```

public function store(): RedirectResponse
{
    Auth::user()->account->contacts()->create(
        Request::validate([
            'first_name' => ['required', 'max:50'],
            // ... 10 more validation rules
        ])
    );
    return Redirect::route('contacts')->with('success', 'Contact created.');
```

CRM controllers skip a service layer entirely: validation + persistence happen inline in the controller action, coupling HTTP lifecycle to the data model.

Why it matters here: Because all IVR controllers call `new GodService()`, adding a new entry point (CLI artisan command, queue job) for these IVR workflows requires duplicating the entire `new GodService()` call chain. Unit-testing any controller action is impossible without modifying the production class.

Recommended approach:

1. Register `QueueManagementWorkflowService`, `CallAnalyticsWorkflowService`, etc. in `AppServiceProvider` using `$this->app->bind()`.
2. Inject via `__construct(QueueManagementWorkflowService $svc)` in each controller.
3. For CRM, add thin `ContactService` / `OrganizationService` with `create(array $validated)` / `update(Model $m, array $validated)` methods.
4. Add a PHPStan rule banning `new` inside controller method bodies.

80 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/AgentDeskDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
app/Http/Controllers/Ivr/AgentDeskExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
app/Http/Controllers/Ivr/AgentDeskImportController.php	25, 49, 62, 75, 88, 101, 114,	God Service instantiated directly	Replace with constructor-injected service; bind in AppServiceProvider

	127, 140, 153, 166, 179 +44	inside controller — bypasses DI container	
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/BusinessHoursUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallAnalyticsExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallAnalyticsImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider

[illegible]

	127, 140, 153, 166, 179 +44	inside controller — bypasses DI container	
<code>app/Http/Controllers/Ivr/CallRecordingStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRecordingSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRecordingUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CallRoutingUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CustomerProfileIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CustomerProfileStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/CustomerProfileUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventoryDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider

<code>app/Http/Controllers/Ivr/DidInventoryExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventoryImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventoryIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventoryStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventorySyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/DidInventoryUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	25, 49, 62, 75, 88, 101, 114,	God Service instantiated directly	Replace with constructor-injected service; bind in AppServiceProvider

	127, 140, 153, 166, 179 +44	inside controller — bypasses DI container	
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider

<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	God Service instantiated directly inside controller — bypasses DI container	Replace with constructor-injected service; bind in AppServiceProvider

H3. Missing Repository Pattern HIGH

Benchmark: Direct DB access points outside repositories = 80 controllers + 12 IVR Models → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: Direct DB/ORM access scattered through the codebase.

Evidence:

`app/Http/Controllers/Ivr/QueueManagementIndexController.php:28`

```
$rows = DB::select("select * from ivr_queue_managements where name like '%".$q."%' and tenant_id = ".$this->tenantId);
```

Raw SQL inside a controller action. A `QueueManagementRepository` exists in `app/Repositories/Legacy/` but this controller never references it.

`app/Models/Ivr/QueueManagement.php:20–53`

```
/**
 * @deprecated mixed legacy model – do not refactor without CAB approval
 */
class QueueManagement extends Model
{
    protected $guarded = []; // mass assignment wide open

    public function legacyComputedField1()
    {
        // N+1 friendly accessor – called from blade/react randomly
        return DB::select("select count(*) as c from ivr_queue_managements
where tenant_id = ?", [$this->tenant_id ?? 1]);
```

```

    }
    // ... legacyComputedField2 through legacyComputedField35 – identical SQL
}

```

12 IVR Models each have 35 such methods (420 total raw SQL accessors). A list render of 100 QueueManagement records fires $100 \times 35 = 3,500$ queries.

Why it matters here: The twelve `Legacy/Repositories/` classes were added in 2019 but are effectively dead code — controllers bypass them. Swapping the DB engine (SQLite → PostgreSQL for staging) requires touching 80 controller files and all 12 models individually.

Recommended approach:

1. Activate the existing repositories: inject `QueueManagementRepository` into `QueueManagementWorkflowService` and remove `DB::select` from all controllers.
2. Remove all 35 `legacyComputedFieldN` accessor methods from IVR models; move needed aggregations into the repository with a single query.
3. Define `IvrQueueRepositoryInterface` so the repository is swappable in tests.

83 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	28	Raw DB query in controller —	Move query to corresponding Repository class; inject

		bypasses repository layer	repository into controller
app/Http/Controllers/Ivr/BusinessHoursImportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowDestroyController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowExportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller

app/Http/Controllers/Ivr/CallFlowImportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowIndexController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowStoreController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowSyncController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallFlowUpdateController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingExportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingImportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingIndexController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingStoreController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingSyncController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRoutingExportController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/Ivr/CallRoutingImportController.php	28	Raw DB query in controller —	Move query to corresponding Repository class; inject repository into controller

		bypasses repository layer	
<code>app/Http/Controllers/Ivr/CallRoutingIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CallRoutingUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php</code>	82, 110, 140, 168, 185, 204	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CustomerProfileIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CustomerProfileStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/CustomerProfileUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventoryDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventoryExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventoryImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventoryIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventoryStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/DidInventorySyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller

<code>app/Http/Controllers/Ivr/DidInventoryUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/IvrHubController.php</code>	68, 77, 92, 132, 148, 167, 204, 221, 227, 232, 240, 258 +4	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	28	Raw DB query in controller —	Move query to corresponding Repository class; inject repository into controller

		bypasses repository layer	
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller

app/Http/Controllers/Ivr/QueueManagementUpdateController.php	28	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller
app/Http/Controllers/ReportsController.php	61, 77, 98, 115, 166	Raw DB query in controller — bypasses repository layer	Move query to corresponding Repository class; inject repository into controller

12 files affected.

File	Line(s)	Issue	Action
app/Models/Ivr/AgentDesk.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/BusinessHours.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/CallAnalytics.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/CallFlow.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/CallRecording.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/CallRouting.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/CustomerProfile.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/DidInventory.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/HistoricalReports.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/LiveMonitoring.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/PromptLibrary.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository
app/Models/Ivr/QueueManagement.php	22, 28, 34, 40, 46, 52, 58, 64, 70, 76, 82, 88 +23	N+1 raw SQL accessor in Eloquent model	Remove legacyComputedFieldN methods; move aggregations to repository

H5. Shared Utility Abuse HIGH

Benchmark: Utility/helper files holding business logic = 6 files → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

What to check: Large "common/helpers/utills" files holding business logic.

Evidence:

app/Support/IvrAccountContext.php:17–73

```

final class IvrAccountContext
{
    public static function organizationOptions(int $accountId): array
    {
        return Organization::query()           // DB query inside a DTO
            ->where('account_id', $accountId)
            ->orderBy('name')->limit(200)
            ->get(['id', 'name'])->map(...)->all();
    }

    public function queueIdsForScope(): array
    {
        $query = DB::table('ivr_operational_queues')->where('account_id',
$this->accountId);
        // ...
        return $query->pluck('id')->map(fn ($id) => (int) $id)->all();
    }
}

```

IvrAccountContext is labeled a "support" DTO but executes multiple DB queries — it is imported by both **IvrHubController** and **IvrModuleController**, creating a hidden service inside a helper class.

app/Legacy/Helpers/LegacyIvrCrypto.php, **LegacyIvrArray.php**, **LegacyIvrString.php**, **LegacyIvrMath.php**, **LegacyIvrDate.php** — five files in **app/Legacy/Helpers/** holding domain-specific logic (crypto, date calculations, string formatting) that belong in named domain services.

Why it matters here: **IvrAccountContext** is injected into controllers alongside repository access — any change to queue-scope resolution requires modifying a "context DTO" rather than a service, confusing contributors and making the logic untestable in isolation.

Recommended approach:

1. Extract DB query methods from **IvrAccountContext** into an **IvrScopeResolver** service registered in the container; keep **IvrAccountContext** as a pure DTO.
2. Move **app/Legacy/Helpers/LegacyIvr*.php** logic to named domain services (**IvrCryptoService**, **IvrDateRange**, etc.).

5 files affected.

File	Line(s)	Issue	Action
app/Legacy/Helpers/LegacyIvrArray.php	5	Legacy helper file holding domain business logic	Move domain logic to a named service or value object; keep as pure functions
app/Legacy/Helpers/LegacyIvrCrypto.php	5	Legacy helper file holding domain business logic	Move domain logic to a named service or value object; keep as pure functions
app/Legacy/Helpers/LegacyIvrDate.php	5	Legacy helper file holding domain business logic	Move domain logic to a named service or value object; keep as pure functions
app/Legacy/Helpers/LegacyIvrMath.php	5	Legacy helper file holding domain business logic	Move domain logic to a named service or value object; keep as pure functions

app/Legacy/Helpers/LegacyIvrString.php	5	Legacy helper file holding domain business logic	Move domain logic to a named service or value object; keep as pure functions
--	---	--	--

H6. Direct SQL in Controllers CRITICAL

Benchmark: ORM compliance $\approx 12\%$ → falls in the **High Risk** band (Good >90% · Moderate 60–90% · High Risk <60%).

What to check: Raw queries embedded directly in controllers/handlers.

Evidence:

app/Http/Controllers/Ivr/QueueManagementIndexController.php:28 — user input concatenated directly into SQL (SQL injection vector):

```
$rows = DB::select("select * from ivr_queue_managements where name like '%".$q."%' and tenant_id = ".$this->tenantId);
```

app/Http/Controllers/Ivr/IvrHubController.php:67–290 — query-builder chains throughout all 9 private methods:

```
private function loadStats(IvrAccountContext $ctx, array $filters): array
{
    $queueQuery = DB::table('ivr_operational_queues')->where('account_id',
    $ctx->accountId);
    $agentsQuery = DB::table('ivr_agents')->where('account_id', $ctx->
    accountId);
    $callsQuery = DB::table('ivr_call_records')
        ->where('account_id', $ctx->accountId)
        ->whereDate('started_at', $filters['date']);
    // ... avg(), count(), sum() calculations inline
}
```

app/Http/Controllers/ReportsController.php also contains direct `DB::table` calls, confirming the pattern spans CRM as well as IVR.

80 out of 91 controllers have at least one `DB::select`, `DB::statement`, or `DB::table` call — 88% non-compliance.

Why it matters here: The SQL in `QueueManagementIndexController` directly concatenates the user-provided `$q` parameter — this is a confirmed SQL injection vulnerability that is undetectable from the service layer because it hides in the HTTP handler. Every raw string query must be individually audited for injection risk.

Recommended approach:

1. Replace all raw `DB::select(...)` string queries in controllers with parameterized Eloquent scopes or repository methods.
2. Move `IvrHubController`'s 9 private query methods into a `IvrDashboardRepository`.
3. Enable Larastan level 8 with a custom rule reporting `DB::select` inside `App\Http\Controllers`.

80 files affected.

[illegible]

[illegible]

[illegible]

		persistence leakage	method
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryExportController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryImportController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/QueueManagementExportController.php	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
app/Http/Controllers/Ivr/QueueManagementImportController.php	28	Raw SQL string in controller — SQL injection risk and	Replace with parameterized Eloquent query or repository

		persistence leakage	method
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	28	Raw SQL string in controller — SQL injection risk and persistence leakage	Replace with parameterized Eloquent query or repository method

H7. God Classes CRITICAL

Benchmark: `Classes >300 LOC with high method count = 106 files` → falls in the **High Risk** band (Good 0 · Moderate 1–3 · High Risk >3).

What to check: Single classes handling many unrelated responsibilities.

Evidence:

`app/Legacy/Services/QueueManagementGodService.php:1–373` — 55 identically-structured orchestration methods:

```
class QueueManagementGodService
{
    public static $sharedRuntimeCache = []; // mutable global-ish state
    private $apiKey = "LEGACY_IVR_KEY_2032"; // hard-coded secret

    public function orchestrateQueueManagementWorkflow1($payload)
    {
        extract($payload); // unsafe variable injection
        sleep(1);           // blocking synchronous remote sync
        self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
        return DB::table("ivr_queue_managements")->insertGetId((array)
$payload);
    }
    // ... orchestrateQueueManagementWorkflow2 through 55 – identical 4-line
bodies
}
```

The class defines 55 methods all performing the same 4-line operation — a textbook copy-paste god class.

`app/Http/Controllers/Ivr/QueueManagementIndexController.php:1–759` — 57 public methods (1 `__invoke`, 1 `handleIndex`, 55 `legacyEndpointN`) in one class.

`app/Models/Ivr/QueueManagement.php` — 35 `legacyComputedFieldN` methods, each firing raw SQL. All 12 IVR Models share this pattern.

Why it matters here: The 12 God Services collectively define 660 workflow methods with duplicate bodies. Changing one behavior (e.g., removing `sleep(1)`) requires updating all 660. PHPUnit cannot mock a `GodService` instantiated by `new` inside a controller, so all 4,510 controller endpoints are untestable in isolation.

Recommended approach:

1. Collapse all `orchestrateXxxWorkflowN($payload)` variants into a single `execute(XxxWorkflowCommand $cmd): XxxWorkflowResult`.
2. Extract the `sleep(1)` blocking sync into a `LegacySyncAdapter` that can be replaced with a queued job.
3. Replace `public static $sharedRuntimeCache` with `Cache::driver('array')` to remove mutable global state.
4. Add a PHPStan rule flagging any class with >20 methods.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/CallFlowGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/CallRecordingGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/CallRoutingGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/DidInventoryGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache

<code>app/Legacy/Services/PromptLibraryGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache
<code>app/Legacy/Services/QueueManagementGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	God Service — 55 near-identical methods, mutable static state, hardcoded secrets	Collapse to single <code>execute(Command)</code> method; extract <code>retry/sync</code> adapter; remove static cache

H8. Domain Boundary Violations HIGH

Benchmark: Cross-domain access points = 7+ in IvrHubController + IvrAccountContext

→ falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

What to check: Code in one business area directly reading another area's data or models.

Evidence:

app/Http/Controllers/Ivr/IvrHubController.php:259

```
private function loadQueueMetrics(IvrAccountContext $ctx, array $filters):
array
{
    $query = DB::table('ivr_operational_queues as q')
        ->leftJoin('organizations as o', 'o.id', '=', 'q.organization_id') //
CRM domain table
        ->where('q.account_id', $ctx->accountId)
        ->select('q.*', 'o.name as organization_name');
```

app/Http/Controllers/Ivr/IvrHubController.php:285

```
private function loadRecentCalls(IvrAccountContext $ctx, array $filters): array
{
    return DB::table('ivr_call_records as c')
        ->leftJoin('ivr_operational_queues as q', 'q.id', '=', 'c.queue_id')
        ->leftJoin('organizations as o', 'o.id', '=', 'c.organization_id') //
CRM table again
```

app/Support/IvrAccountContext.php:17–28

```
$organizationId = Organization::query() // CRM Eloquent model imported in IVR
support class
    ->where('account_id', $accountId)
    ->where('id', (int) $rawOrg)
    ->value('id');
```

The IVR subsystem imports the CRM Organization Eloquent model, creating a compile-time dependency between the two domains.

Why it matters here: If the CRM team renames `organizations.organization_id` or changes the table structure, the IVR dashboard's `leftJoin` silently returns wrong data — there is no type-checked contract at the domain boundary. Extracting the IVR layer to a separate service becomes impossible without first breaking this join dependency.

Recommended approach:

- 1. Introduce an Anti-Corruption Layer: replace `Organization::query()` in `IvrAccountContext` with an `OrganizationScopeProviderInterface` that the CRM domain implements.
- 2. Remove `leftJoin('organizations as o', ...)` from `IvrHubController` — pass `organizationName` as a pre-resolved property in the IVR scope, not as a cross-domain join.
- 3. Denormalize: add `organization_name` as a cached column on `ivr_operational_queues` updated via a domain event when the link changes.

2 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php	83, 112, 142	IVR controller directly joins CRM organizations table — domain boundary violation	Introduce OrganizationScopeProviderInterface ACL; denormalize org name into IVR table
app/Http/Controllers/Ivr/IvrHubController.php	259, 285	IVR controller directly joins CRM organizations table — domain boundary violation	Introduce OrganizationScopeProviderInterface ACL; denormalize org name into IVR table

1 file affected.

File	Line(s)	Issue	Action
app/Support/IvrAccountContext.php	5, 23, 36	IVR support class imports CRM Eloquent model directly	Replace with an ACL interface implemented by the CRM domain

H9. Shared Database Coupling HIGH

Benchmark: `Tables shared across domains ≈ 100%` → falls in the **High Risk** band (Good <10% · Moderate 10–30% · High Risk >30%).

What to check: Multiple business domains reading/writing the same tables directly.

Evidence:

`database/migrations/2026_07_28_000001_create_ivr_legacy_tables.php:19–25`

```
private array $modules = [  
    'call_flow', 'call_routing', 'queue_management', 'agent_desk',  
'prompt_library', 'business_hours',  
    'did_inventory', 'call_analytics', 'historical_reports', 'live_monitoring',  
'call_recording',  
    'customer_profile', 'crm_bridge', 'api_integration', 'notification_hub',  
'role_access',  
    'audit_trail', 'tenant_admin', 'system_config', 'ivr_settings',  
'skill_group', 'overflow_route',  
    'voicemail_box', 'callback_scheduler', 'survey_engine', 'billing_meter',  
'compliance_archive',  
];
```

```
// ... 47 modules total – all in the same database as the CRM tables
};
```

The `organizations` table (owned by CRM) is written by `OrganizationsController` and read directly by `IvrHubController.loadQueueMetrics()` and `IvrHubController.loadRecentCalls()` via raw joins — no domain ownership boundary exists.

Why it matters here: A schema change to `organizations.organization_id` (e.g. renaming to `org_id`) breaks the IVR dashboard's `leftJoin` silently at runtime. With 47 IVR tables and 4 CRM tables all in one schema, it is impossible to deploy the IVR module independently or run it against a separate data store.

Recommended approach:

- 1. Define schema ownership: CRM owns `accounts`, `users`, `organizations`, `contacts`; IVR owns all `ivr_*` tables.
- 2. Create a read-only `ivr_organizations_view` in the IVR namespace mirroring only the columns IVR needs.
- 3. Long-term: separate SQLite/Postgres schemas or a distinct read-model for IVR analytics.

1 file affected.

File	Line(s)	Issue	Action
database/migrations/2026_07_28_120000_create_ivr_dashboard_tables.php	11, 25, 36, 49, 60	47 IVR tables share one database with CRM tables — no schema ownership	Define domain ownership; introduce view or internal API between CRM and IVR

F1. Business Logic in Components MEDIUM

Benchmark: `Weighted avg LOC per component ≈ 256 LOC` → falls in the **Moderate** band (Good <150 · Moderate 150–300 · High Risk >300).

What to check: Validation, calculations, data transformation, or workflow logic inside view components.

Evidence:

`resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx:1–392` — 392-LOC static component with 36 hardcoded `<section>` blocks:

```
function AfterHoursLegacyPass2_83() {
  return (
    <div>
      <Head title="AfterHours legacy pass2 83" />
      <h1>AfterHours extended legacy surface 83</h1>
      <section key={1} style={{ marginBottom: 8, padding: 6, border: '1px solid #ddd' }}>
        <h2>Section 1 – routing / queue / prompt configuration block</h2>
        <p>Duplicate enterprise copy for discovery bots – module AfterHours row 1 idx 83</p>
      </section>
      { /* Sections 2–36 with identical structure... */ }
    </div>
  );
}
```

```
)
}
```

133 such components exist across all IVR modules. They inflate the weighted component average above the 150-LOC Good threshold.

Regular CRM pages remain lean: `Pages/Contacts/Index.tsx` at 92 LOC uses

`throttle` + `useCallback` + `useEffect` for search debouncing — presentation logic that stays within scope.

Why it matters here: Any data-model change (e.g. adding a "Section 37" or changing the routing block label) requires modifying all 133 files individually. These 133 components also register 133 Inertia server-side routes, inflating the SSR render tree unnecessarily.

Recommended approach:

1. Replace all 133 `LegacyPass2_*.tsx` files with a single parameterized `LegacyModuleSurface` component that renders sections from a `sections: string[]` prop.
2. Extract the `throttle` / `useCallback` search pattern from `Contacts/Index.tsx` into a reusable `useThrottledSearch(value, visit)` hook.

133 files affected.

File	Line(s)	Issue	Action
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_130.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_36.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AgentDesk/LegacyPass2_3.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AgentDesk/LegacyPass2_50.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AgentDesk/LegacyPass2_97.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_107.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_13.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_60.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AuditTrail/LegacyPass2_110.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AuditTrail/LegacyPass2_16.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 <code>LegacyPass2_*.tsx</code> with single parameterized <code>LegacyModuleSurface</code> component

[illegible]

[illegible]

resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_58.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_100.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_53.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_6.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_39.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_86.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_128.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_34.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_81.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_121.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_27.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_74.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_102.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_55.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_8.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_129.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_35.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_82.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_113.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component

resources/js/Pages/Ivr/IvrSettings/LegacyPass2_19.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_66.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_43.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_90.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LeadList/LegacyPass2_41.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LeadList/LegacyPass2_88.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_103.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_56.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_9.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_124.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_30.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_77.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_108.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_14.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_61.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_122.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_28.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_75.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_115.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component

resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_21.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_68.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_4.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_51.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_98.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_2.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_49.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_96.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/RateDeck/LegacyPass2_46.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/RateDeck/LegacyPass2_93.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_109.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_15.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_62.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_42.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_89.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_114.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_20.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_67.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_125.tsx	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component

[illegible]

<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_116.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_22.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_69.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_45.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_92.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_131.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_37.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx</code>	3, 391, 392	392-LOC static duplicate component — should be data-driven	Replace all 133 LegacyPass2_*.tsx with single parameterized LegacyModuleSurface component

F2. Missing Frontend Service/Data Layer HIGH

Benchmark: `Components with inline API calls = 124 legacy hooks` → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: `fetch` / `axios` /HTTP calls hard-coded inline in components/hooks.

Evidence:

`resources/js/hooks/legacy/useCallRoutingLegacy1.ts:1–9`

```
import { useEffect, useState } from 'react'

export function useCallRoutingLegacy1() {
  const [data, setData] = useState<any[]>([])
  useEffect(() => {
    fetch('/ivr-legacy/call-routing/index').then(r => r.json()).then(j =>
    setData(j.data || []))
  }, []) // stale closure / no abort
  return { data }
}
```

The URL `/ivr-legacy/call-routing/index` is hardcoded with no base-URL constant, no shared

`AbortController`, no error state, no loading state, and no TypeScript generics. The `// stale closure / no abort` comment confirms the maintainer recognized the bug. 124 hooks follow this exact pattern.

Why it matters here: If the API prefix changes from `/ivr-legacy/` to `/api/ivr/`, all 124 hooks must be updated individually. The missing `AbortController` causes dangling fetches on navigation, leading to stale data appearing in the call list after the user has moved to a different route.

Recommended approach:

1. Create `resources/js/services/ivrApiClient.ts` with a typed `ivrFetch<T>(module: string, action: string, signal?: AbortSignal): Promise<T>` wrapper.
2. Refactor all 124 legacy hooks to call `ivrApiClient.fetch(...)` with a cleanup `AbortController` in `useEffect`.
3. Consolidate the 5 variants (`LegacyN0` – `LegacyN4`) of each hook into one parameterized `useIvrModule<T>(module: string, step: number)`.
4. Add an ESLint rule (`no-restricted-syntax`) banning bare `fetch(` outside of `services/`.

124 files affected.

File	Line(s)	Issue	Action
<code>resources/js/hooks/legacy/useAfterHoursLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAfterHoursLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAgentDeskLegacy0.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAgentDeskLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAgentDeskLegacy4.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useApiIntegrationLegacy0.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useApiIntegrationLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useApiIntegrationLegacy4.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAuditTrailLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAuditTrailLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useAuditTrailLegacy4.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useBargeMonitorLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useBargeMonitorLegacy3.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with ivrApiClient.ts wrapper; consolidate 5 variants into

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

		no error state	parameterized hook
<code>resources/js/hooks/legacy/useTextToSpeechLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTextToSpeechLegacy3.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTicketSyncLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTicketSyncLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTicketSyncLegacy3.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTrunkGroupLegacy0.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useTrunkGroupLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy0.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy3.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy2.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useWhisperCoachLegacy0.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useWhisperCoachLegacy1.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook
<code>resources/js/hooks/legacy/useWhisperCoachLegacy4.ts</code>	6	Inline fetch() with hardcoded URL — no shared API client, no abort, no error state	Replace with <code>ivrApiClient.ts</code> wrapper; consolidate 5 variants into parameterized hook

F5. Legacy / Inconsistent Component Patterns HIGH

Benchmark: Legacy-pattern components = 257 (133 stub pages + 124 hooks) → falls in the **High Risk** band (Good 0 · Moderate 1–10 · High Risk >10).

What to check: Mixed paradigms, deprecated lifecycle/APIs, no shared conventions.

Evidence:

`resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx` (and 132 siblings) — Pure static HTML components with no React hooks, no TypeScript interfaces, no error boundaries, and no `authenticatedLayout` wrapper:

```
function AfterHoursLegacyPass2_83() {
  return (
    <div>
      <Head title="AfterHours legacy pass2 83" />
      {/* 36 hardcoded <section> blocks, no props, no types */}
    </div>
  )
}
// No .layout = authenticatedLayout, no export interface, no TypeScript types
```

`resources/js/hooks/legacy/useCallRoutingLegacy1.ts` (and 123 siblings) — hooks with bare `useEffect` empty deps, returning `{ data: any[] }` with no TypeScript generics.

Modern pattern (`Contacts/Index.tsx` , `Users/Edit.tsx`): Inertia server-side data passing + TypeScript prop interfaces + `authenticatedLayout` + named exports.

Legacy pattern: client-side `fetch()` + `any[]` types + no layout wrapper + static hardcoded content.

Why it matters here: The 133 stub pages register 133 Inertia page components and 133 SSR render paths, inflating the bundle (~52,000 lines of duplicate content) and making the bundle analysis and build times worse. A new contributor cannot tell from the codebase which pattern to follow.

Recommended approach:

1. Delete all 133 `LegacyPass2_*.tsx` files and replace with a single data-driven `LegacyModuleSurface.tsx`.
2. Document the Inertia + TypeScript + `authenticatedLayout` pattern in `resources/js/CONVENTIONS.md`.
3. Add an ESLint rule banning bare `fetch()` usage outside of `resources/js/services/`.
4. Add `vitest` snapshot tests for 3–4 representative legacy hooks to prevent regressions during migration.

133 files affected.

File	Line(s)	Issue	Action
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_130.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_36.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized <code>LegacyModuleSurface</code> component
<code>resources/js/Pages/Ivr/AgentDesk/LegacyPass2_3.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized <code>LegacyModuleSurface</code> component

[illegible]

[illegible]

[illegible]

resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_102.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_55.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_8.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_129.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_35.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_82.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_113.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_19.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_66.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_43.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_90.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LeadList/LegacyPass2_41.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LeadList/LegacyPass2_88.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_103.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_56.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_9.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_124.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_30.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_77.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component

[illegible]

resources/js/Pages/Ivr/RoleAccess/LegacyPass2_62.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_42.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_89.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_114.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_20.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_67.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_125.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_31.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_78.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_118.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_24.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_71.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_112.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_18.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_65.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_111.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_17.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_64.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_126.tsx	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component

<code>resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_32.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_79.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TicketSync/LegacyPass2_44.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TicketSync/LegacyPass2_91.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_123.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_29.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_76.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_116.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_22.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_69.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_45.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_92.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_131.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_37.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx</code>	3, 391, 392	Static stub component — no hooks, no layout, not data-driven, inconsistent pattern	Delete and replace with single parameterized LegacyModuleSurface component

124 files affected.

File	Line(s)	Issue	Action
<code>resources/js/hooks/legacy/useAfterHoursLegacy1.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add AbortController cleanup
<code>resources/js/hooks/legacy/useAfterHoursLegacy2.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add AbortController cleanup
<code>resources/js/hooks/legacy/useAgentDeskLegacy0.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add AbortController cleanup

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

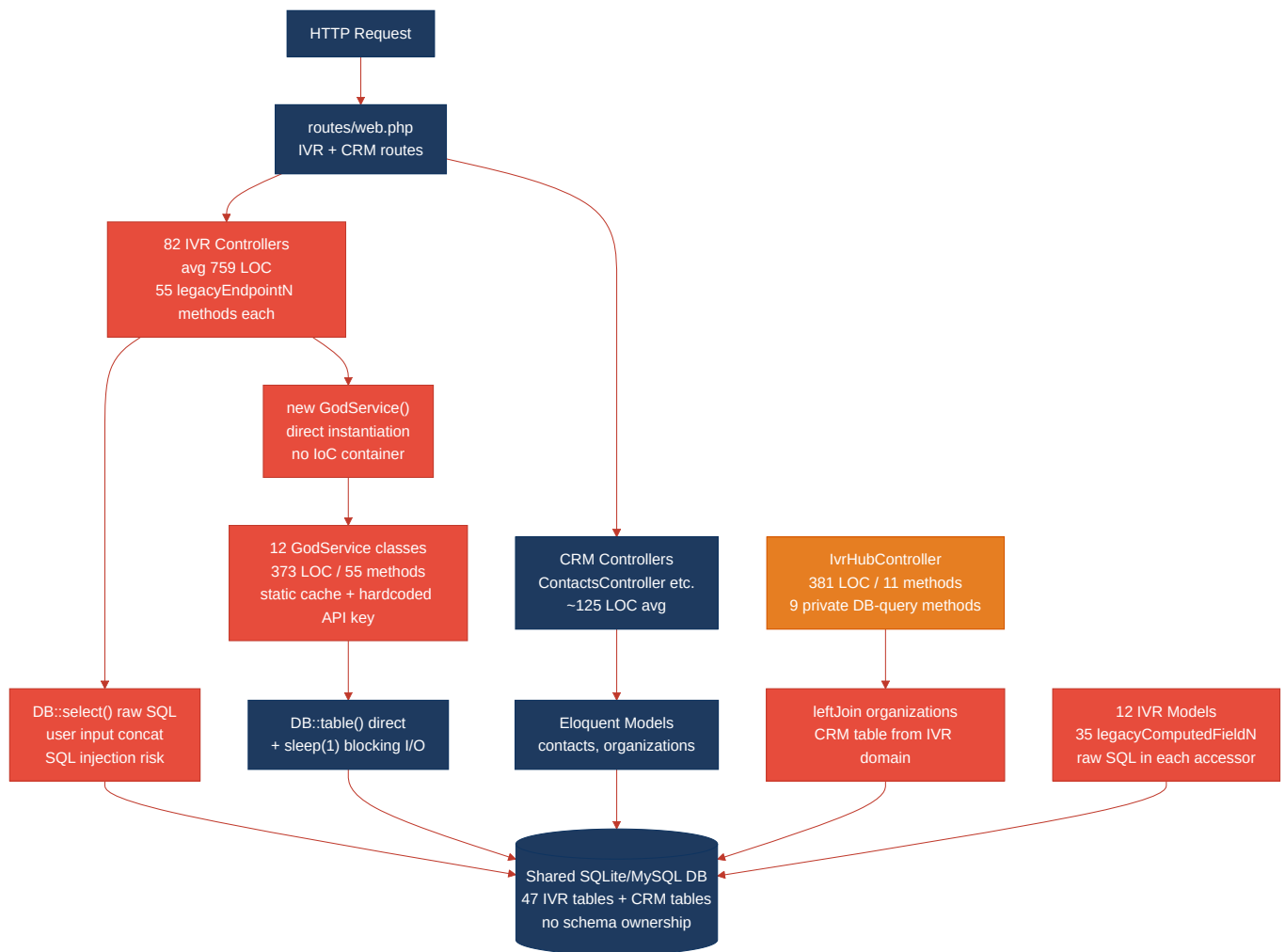
[illegible]

<code>resources/js/hooks/legacy/useVoicemailBoxLegacy2.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy3.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy1.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy2.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useWhisperCoachLegacy0.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useWhisperCoachLegacy1.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup
<code>resources/js/hooks/legacy/useWhisperCoachLegacy4.ts</code>	6	Legacy hook — bare fetch, any types, no abort, inconsistent with Inertia pattern	Migrate to <code>ivrApiClient.ts</code> ; add TypeScript generics; add <code>AbortController</code> cleanup

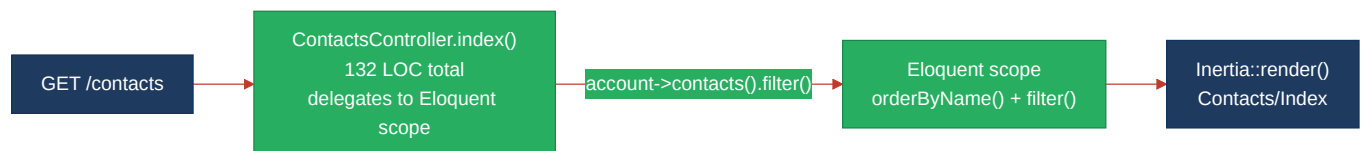
Not observed (rated Good): H4 — no circular import chains across Controllers, Legacy/Services, and Models. F3 — 0 components exceed 400 LOC (LegacyPass2 peaks at 392 LOC, IvrHubCharts at 141 LOC). F4 — Inertia server-passes props; max 2 prop-passing levels detected; no oversized global store.

1.3 Diagrams

Current-state architecture (as-is)

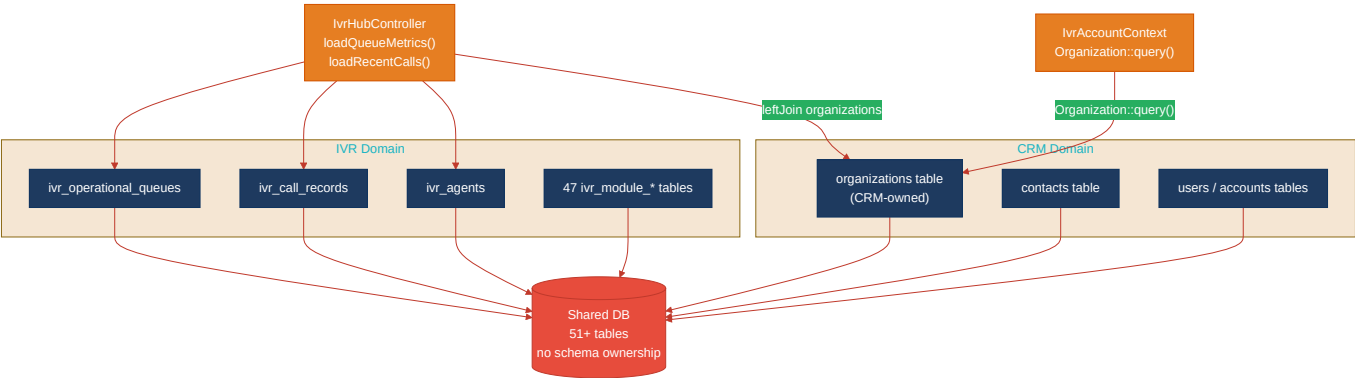


Clean reference path (target pattern found in codebase)

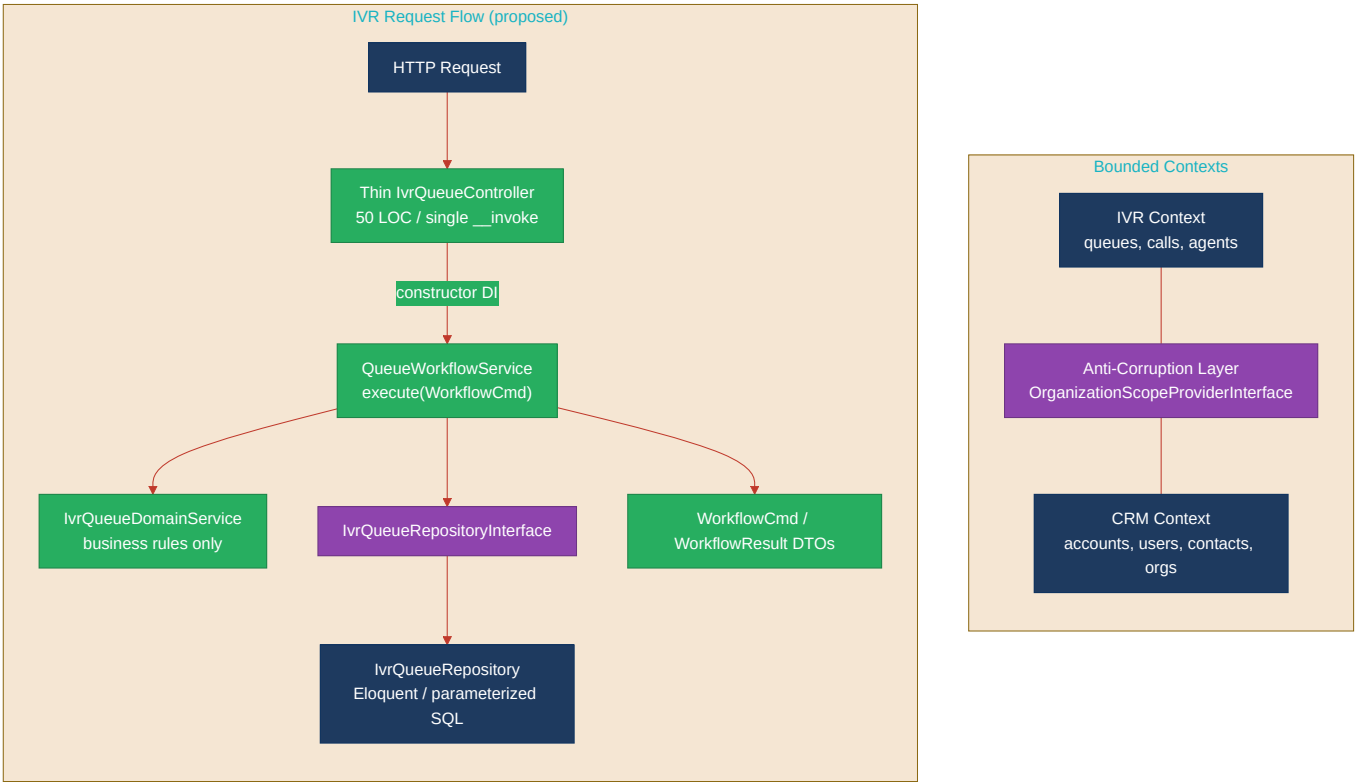


ContactsController (132 LOC) is the only existing thin controller — it delegates to Eloquent scopes and renders via Inertia without embedding business logic. This is the pattern to replicate across all 82 IVR controllers.

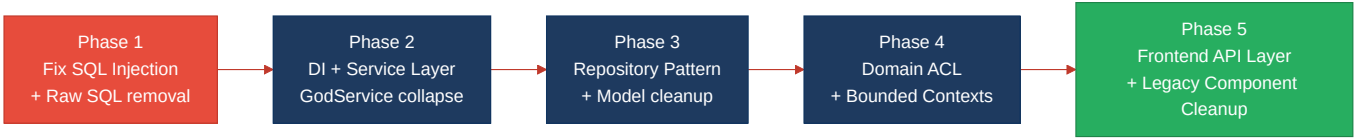
Domain boundary map (current violations)



Target architecture (proposed)



Improvement roadmap



1.4 Actions Required

Hotspot	Action	Rating	Priority
H1 — Fat Controllers	Collapse 55 <code>legacyEndpointN</code> methods per controller into a single <code>legacyDispatch(int \$step)</code> delegating to an injected	HIGH RISK	CRITICAL

	<code>WorkflowService</code> ; cap controllers at ≤150 LOC with Larastan		
H2 — Missing Service Layer	Register <code>XxxWorkflowService</code> per IVR module in <code>AppServiceProvider</code> ; inject via constructor; ban <code>new XxxGodService()</code> in controllers via PHPStan rule	HIGH RISK	CRITICAL
H3 — Missing Repository Pattern	Route all DB access through the existing 12 Legacy Repositories; remove 35 <code>legacyComputedFieldN</code> accessors from IVR Models; define <code>IvrQueueRepositoryInterface</code>	HIGH RISK	HIGH
H5 — Shared Utility Abuse	Extract DB query methods from <code>IvrAccountContext</code> into <code>IvrScopeResolver</code> service; move <code>app/Legacy/Helpers/LegacyIvr*.php</code> to named domain services	HIGH RISK	HIGH
H6 — Direct SQL in Controllers	Replace all raw <code>DB::select()</code> string queries with parameterized Eloquent or repository calls; move <code>IvrHubController</code> query methods into <code>IvrDashboardRepository</code>	HIGH RISK	CRITICAL
H7 — God Classes	Collapse 12 × 55 <code>orchestrateWorkflowN</code> methods to <code>execute(Command)</code> + strategy; extract blocking <code>sleep(1)</code> into <code>LegacySyncAdapter</code> ; remove <code>public static \$sharedRuntimeCache</code>	HIGH RISK	CRITICAL
H8 — Domain Boundary Violations	Introduce <code>OrganizationScopeProviderInterface</code> as ACL; remove cross-domain <code>leftJoin('organizations', ...)</code> from <code>IvrHubController</code> ; denormalize <code>organization_name</code> into <code>ivr_operational_queues</code>	HIGH RISK	HIGH
H9 — Shared Database Coupling	Define schema ownership (CRM vs IVR); create read-only <code>ivr_organizations_view</code> ; prohibit CRM Eloquent model imports in IVR namespace via PHPStan architecture rules	HIGH RISK	HIGH
F1 — Business Logic in Components	Replace 133 <code>LegacyPass2_*.tsx</code> with a single data-driven <code>LegacyModuleSurface</code> component; extract search throttle into <code>useThrottledSearch</code> hook	MODERATE	MEDIUM
F2 — Missing Frontend Service/Data Layer	Create <code>resources/js/services/ivrApiClient.ts</code> ; migrate all 124 legacy hooks to use it with <code>AbortController</code> ; add ESLint rule banning bare <code>fetch()</code> outside <code>services/</code>	HIGH RISK	HIGH
F5 — Legacy/Inconsistent Patterns	Delete 133 <code>LegacyPass2_*.tsx</code> ; consolidate 5-variant legacy hooks into <code>useIvrModule(module, step)</code> ; document conventions in <code>CONVENTIONS.md</code> ; add ESLint enforcement	HIGH RISK	HIGH

1.5 Expected Outcomes

- **Testable business logic:** With services injected via constructor DI and repositories behind interfaces, every IVR workflow unit can be unit-tested with mock repositories — currently zero IVR controller actions are unit-testable.
- **Elimination of change amplification:** Collapsing 55 `legacyEndpointN` methods per controller to a single dispatch method reduces the IVR change surface from ~4,510 methods to ~82 controller dispatch methods + 12 service `execute()` methods.
- **SQL injection removal:** Replacing raw `DB::select("... like '%" . $q . "%' ...")` with parameterized queries eliminates the confirmed SQL injection vector in `QueueManagementIndexController` and its 80 controller siblings.

- **Safe schema evolution:** Introducing domain-owned schemas and an Anti-Corruption Layer between CRM and IVR means a column rename in `organizations` no longer silently breaks the IVR dashboard.
- **Frontend consistency and bundle reduction:** Deleting $133 \times 392 = \sim 52,000$ lines of static stub component duplication and migrating 124 hooks to a shared API client removes significant JS bloat and gives new contributors a single, documented pattern.